



Unit-III: Unsupervised Learning (AY-2026-27)



3.1

Evaluating Machine Learning algorithms and Model Selection





Introduction

- Why evaluation is important in ML?
- Difference between training performance and real-world performance
- Bias–variance tradeoff

2. Performance Metrics

□ For Classification

- Accuracy, Precision, Recall, F1-score
- Confusion matrix
- ROC curve, AUC

□ For Regression

- Mean Squared Error (MSE), Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- R^2 score (coefficient of determination)

□ For Ranking/Recommendation

- Precision@k, Recall@k, MAP, NDCG



3. Validation Techniques

- ❑ Hold-out method (train/test split)
- ❑ k-Fold Cross-Validation
- ❑ Leave-One-Out Cross-Validation (LOOCV)
- ❑ Stratified sampling (for imbalanced datasets)



4. Model Selection Strategies

- ❑ Comparing multiple algorithms
- ❑ Hyperparameter tuning
 - ❑ Grid Search
 - ❑ Random Search
 - ❑ Bayesian Optimization
- ❑ Automated Machine Learning (AutoML)

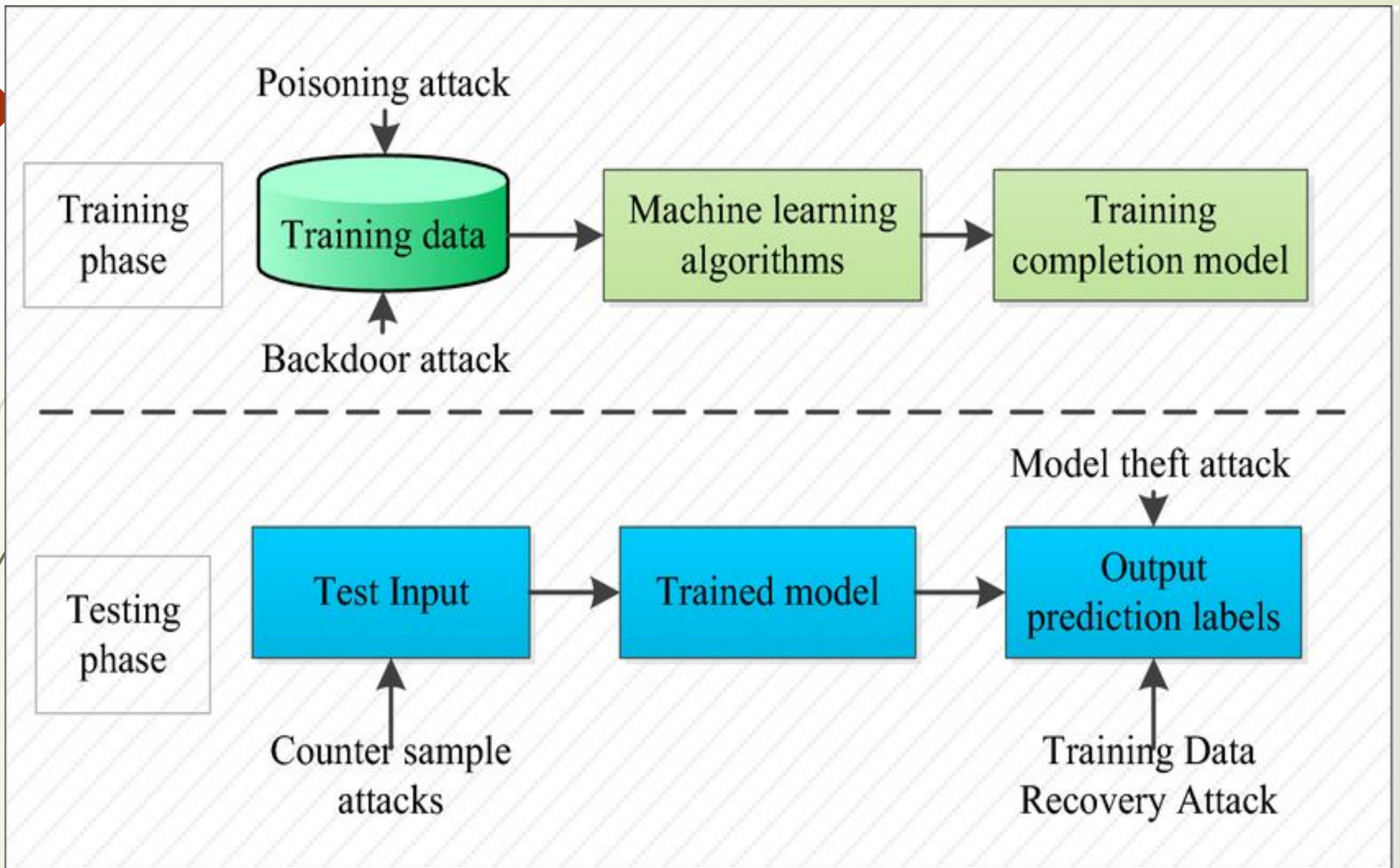
5. Overfitting and Underfitting

- Causes and detection
- Regularization (L1, L2, dropout)
- Early stopping



6. Model Complexity and Generalization

- Bias–variance tradeoff revisited
 - Model capacity vs dataset size
 - Occam's razor principle in ML
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3.2

**Introduction to
Statistical Learning
Theory**

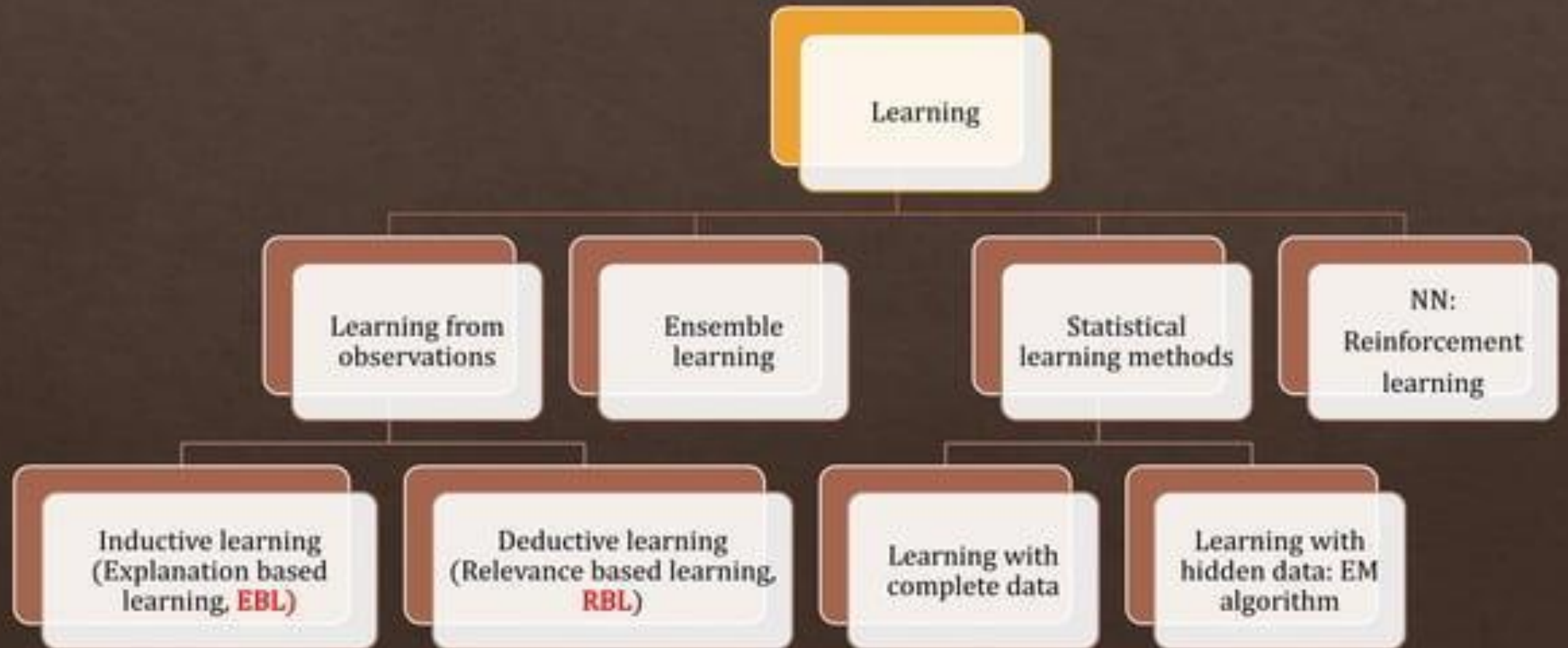
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- Statistical Learning Theory (SLT) is a theoretical framework for understanding **how machines learn from data** and make predictions.
 - It provides the **mathematical foundations** behind many modern machine learning algorithms and helps explain their performance, limitations, and generalization ability.

Statistical Learning Theory

- The goals of learning are **understanding and prediction**.
- Learning falls into many categories i.e. Supervised, unsupervised, online, reinforcement learning.
- The learning problem consists of **inferring the function that maps between the input and the output**, such that the learning function can be used to predict output from future input.



The place of Statistical Learning Methods in LEARNING





□ Learning Problem

- Input: A dataset consisting of feature vectors and corresponding labels.
- Goal: Learn a function (hypothesis) that maps inputs to outputs with minimal error.

□ Hypothesis Space (H)

- The set of all possible models the learning algorithm can choose from.
- Example: In linear regression, the hypothesis space is the set of all linear functions.

• Risk and Loss Functions

- **Loss function:** Measures the error of predictions (e.g., squared error, hinge loss, cross-entropy).
- **Expected Risk (True Risk):** Average loss over the entire data distribution (unknown in practice).
- **Empirical Risk:** Average loss over the training dataset (known).
- **Empirical Risk Minimization (ERM)**
A principle where the learner chooses the hypothesis that minimizes the training error.
- **Problem:** ERM may lead to **overfitting** if the hypothesis is too complex.

• Generalization

- The ability of a model to perform well on **unseen data**.
- Statistical Learning Theory provides tools (like VC dimension, bounds, and
- regularization) to study generalization

• VC Dimension and Capacity Control

- **Vapnik–Chervonenkis (VC) dimension**: A measure of the complexity or capacity of a hypothesis space.
- Models with very high VC dimension can overfit; too low VC dimension may underfit.



•Structural Risk Minimization (SRM)

- An approach that balances **model complexity** and **training error**.
- Introduces **regularization** to avoid overfitting and improve generalization.

•PAC Learning (Probably Approximately Correct)

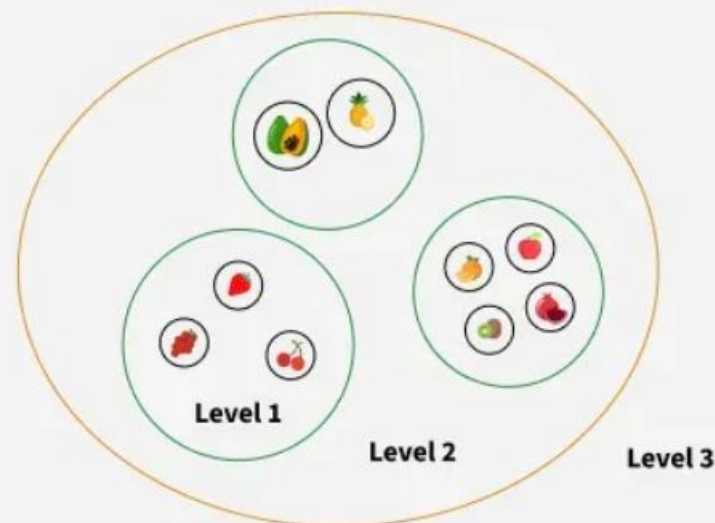
- Framework that defines conditions under which a learning algorithm can guarantee
- that the learned hypothesis is close to the true function, with high probability.

Hierarchical Clustering in Machine Learning

- ❑ **Hierarchical clustering** is an **unsupervised machine learning** technique used to group similar data points into clusters.
- ❑ Unlike algorithms such as K-Means, it does **not require the number of clusters to be specified in advance**.

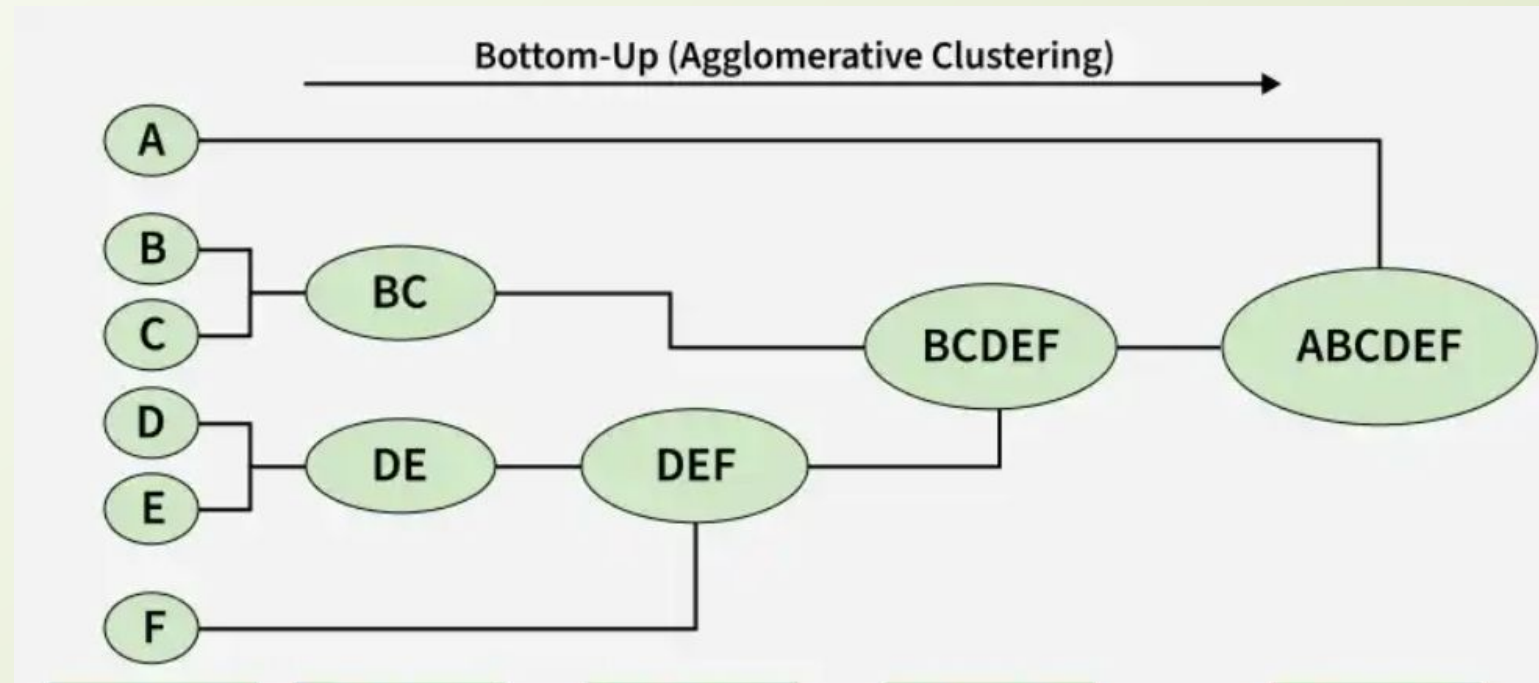
What is Hierarchical Clustering

Hierarchical clustering is an unsupervised machine learning algorithm that groups data into a tree of nested clusters



Agglomerative Hierarchical Clustering (Bottom-Up)

- Starts with each data point as an individual cluster.
- Repeatedly merges the two most similar clusters.
- Continues until all points belong to a single cluster or a stopping criterion is reached.
- This is the most commonly used approach.



Hierarchical Clustering

Hierarchical clustering is an **unsupervised learning** technique that builds a hierarchy of clusters.

Agglomerative (Bottom-Up) Approach

Each point starts as a separate cluster.
The two most similar clusters are merged repeatedly until only one cluster remains.

Step 1

Initial clusters



Step 2

Merge closest clusters (A and B)



Step 3

Merge next closest clusters (D and E)



Step 4

Merge next closest cluster ((A,B) and C)



Step 5

Merge the last two clusters

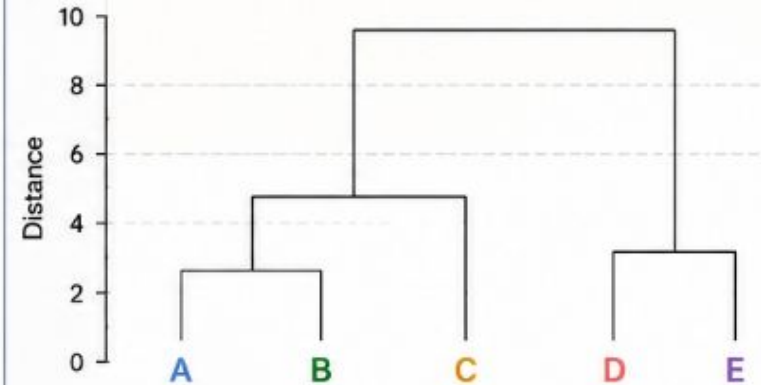


Step 6

Single cluster

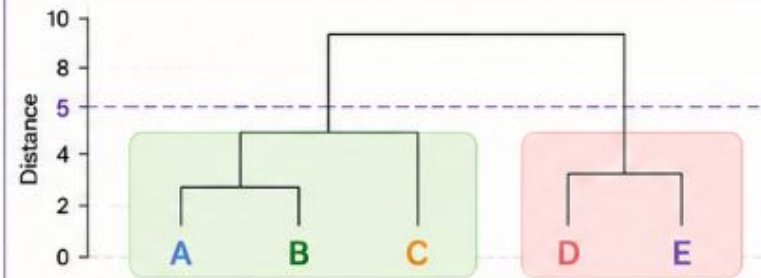


Dendrogram (Result)



- The height at which clusters are merged is shown by the vertical axis (distance).
- Cut the dendrogram at a chosen height to get the desired number of clusters.

Clusters (Example: Cut at distance = 5)



Two clusters are formed: {A, B, C} and {D, E}

Key Points:

- No need to specify number of clusters in advance
- Produces a hierarchy of clusters
- Visualized using a dendrogram



Agglomerative hierarchical clustering

1. **How Agglomerative Hierarchical Clustering Works**
 2. Suppose you have five data points: A, B, C, D, E.
 3. Treat each point as a separate cluster.
 4. Compute the distance between all clusters.
 5. Merge the two closest clusters.
 6. Recalculate distances.
 7. Repeat until only one cluster remains.
- 



Linkage Methods

Linkage Method

Single Linkage

Complete Linkage

Average Linkage

Ward's Linkage

Description

Minimum distance between two clusters

Maximum distance between two clusters

Average distance between all pairs of points

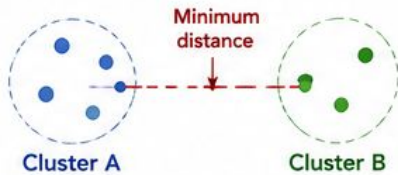
Minimizes the increase in within-cluster variance

Linkage Methods in Hierarchical Clustering

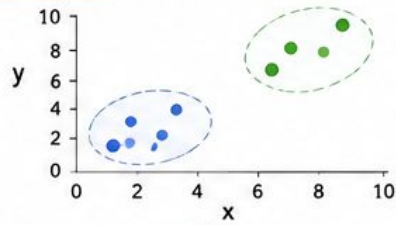
Linkage method defines how the distance between two clusters is computed.

1. Single Linkage (Minimum Linkage)

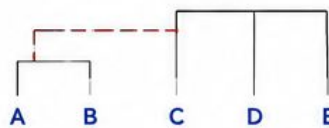
Distance between two clusters is the **minimum** distance between any pair of points, one from each cluster.



Example



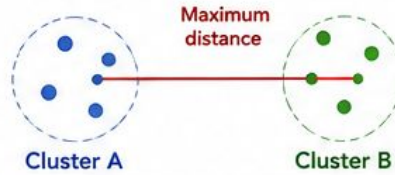
Dendrogram



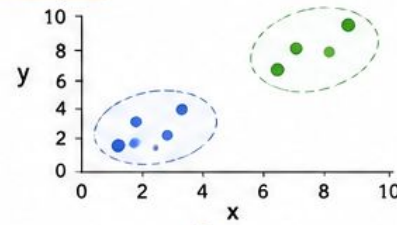
Produces long, chain-like clusters.
Sensitive to noise and outliers.

2. Complete Linkage (Maximum Linkage)

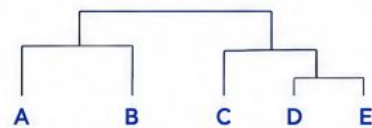
Distance between two clusters is the **maximum** distance between any pair of points, one from each cluster.



Example



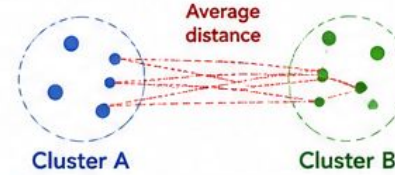
Dendrogram



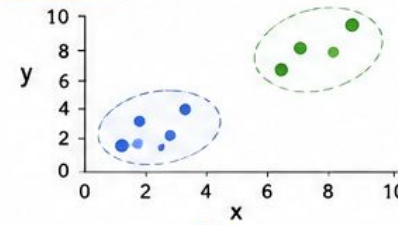
Produces compact, spherical clusters.
Less sensitive to noise and outliers.

3. Average Linkage (UPGMA)

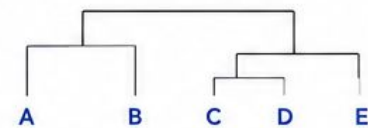
Distance between two clusters is the **average** distance between all pairs of points, one from each cluster.



Example



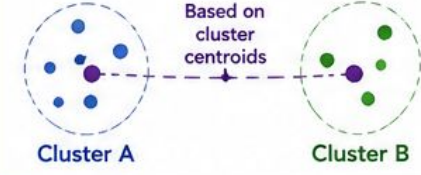
Dendrogram



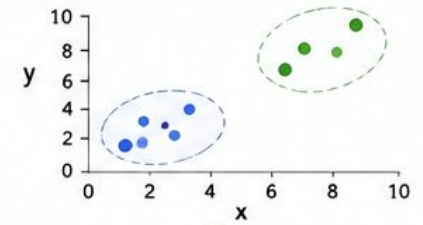
Balances between single and complete linkage. Moderate clusters.

4. Ward's Linkage (Minimum Variance)

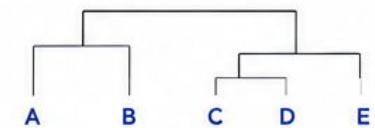
Merges clusters so that the increase in total **within-cluster variance** is minimum.



Example



Dendrogram



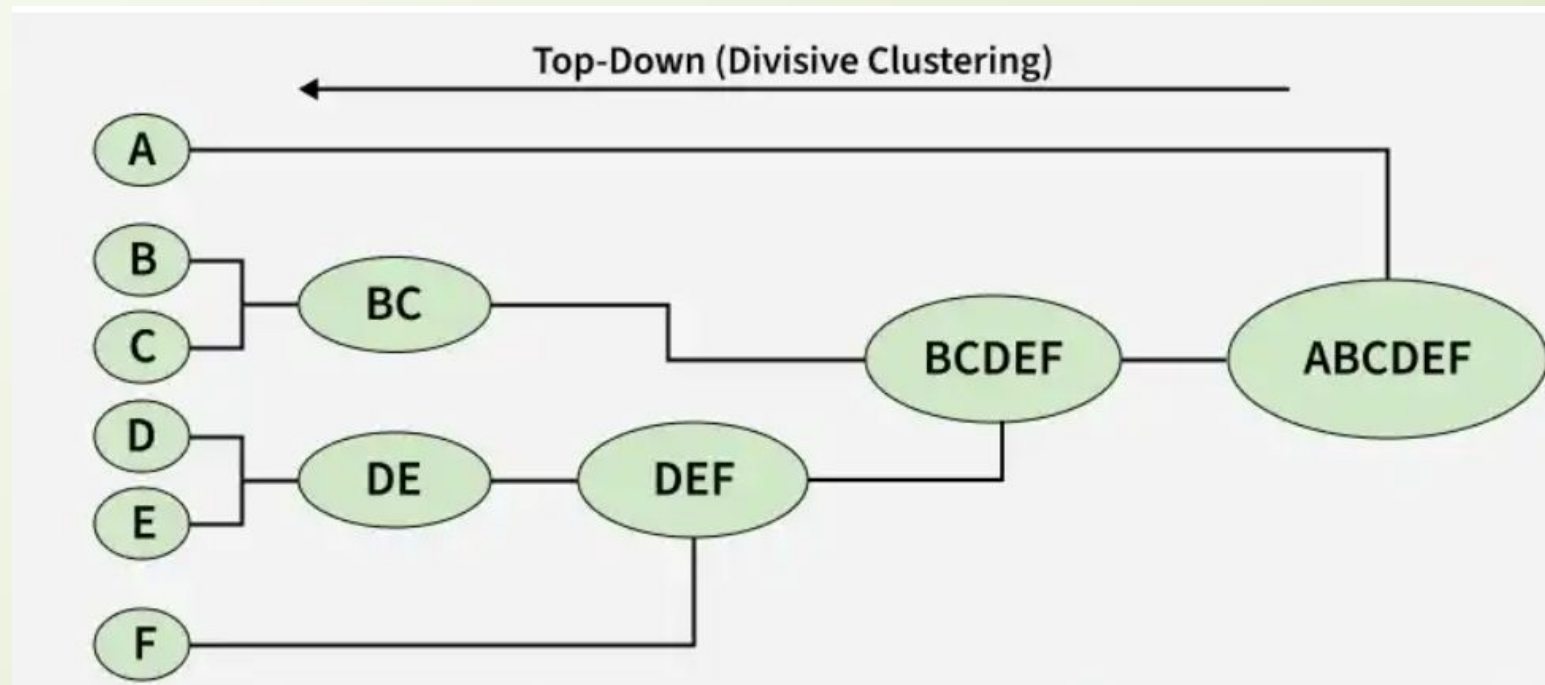
Produces clusters with minimum variance. Generally good results.

Summary Table

Linkage Method	Distance Between Clusters	Cluster Shape	Robustness to Noise	Best Used When
Single (Minimum)	Minimum distance between points	Long, irregular	Low	Clusters are non-compact / chaining allowed
Complete (Maximum)	Maximum distance between points	Compact, spherical	High	Clusters are compact and well-separated
Average (UPGMA)	Average distance between all pairs	Moderate	Medium	General purpose
Ward's (Minimum Variance)	Minimize increase in within-cluster variance	Compact, spherical	High	Clusters of similar size and variance

Divisive Hierarchical Clustering (Top-Down)

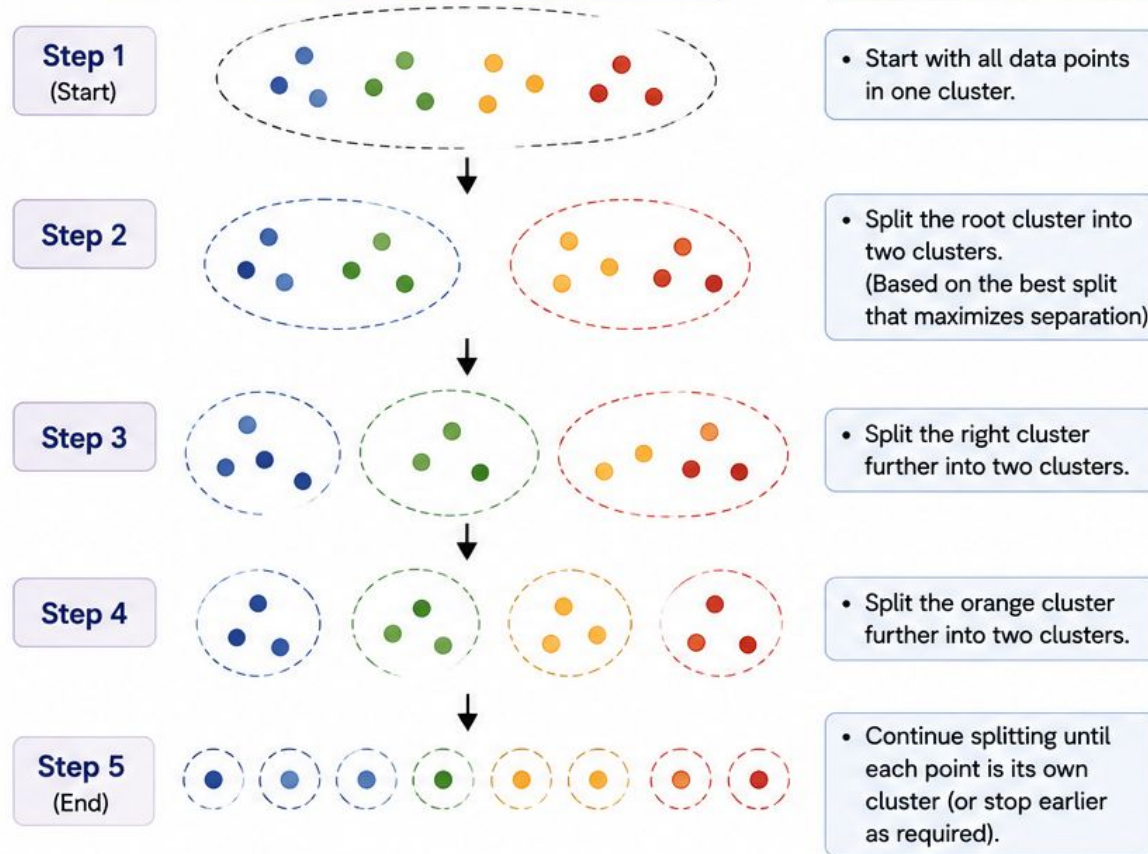
- Starts with all data points in one cluster.
- Repeatedly splits clusters into smaller clusters.
- Continues until each data point forms its own cluster or the desired number of clusters is obtained.



Divisive Hierarchical Clustering (Top-Down Approach)

Divisive hierarchical clustering starts with all data points in a single cluster and recursively splits clusters into smaller clusters until each point forms its own cluster (or until a stopping criterion is met).

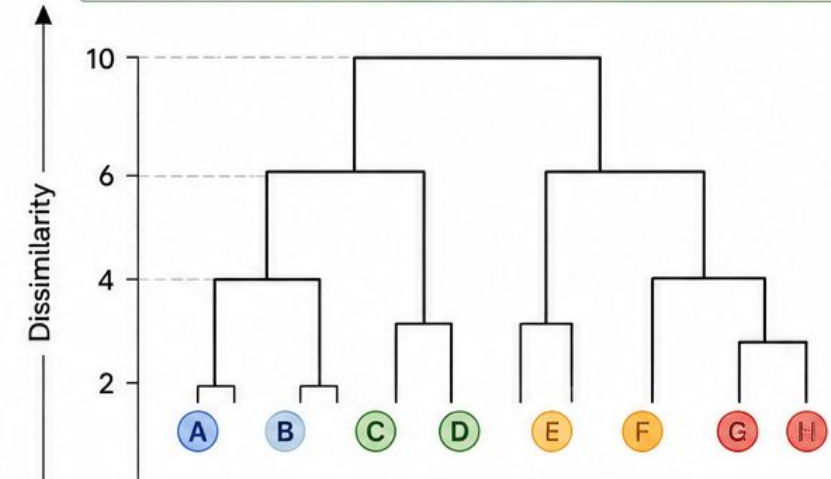
How it works (Example)



Key Points

- Top-Down approach (opposite of agglomerative).
- At each step, choose the cluster to split and the best way to split it.
- Stopping criteria: number of clusters desired / minimum cluster size / maximum tree depth.

Dendrogram (Divisive)



Cutting the dendrogram at the dashed line (e.g., at dissimilarity = 6) gives 2 clusters: {A, B, C, D} and {E, F, G, H}

Divisive vs. Agglomerative

	Divisive (Top-Down)	Agglomerative (Bottom-Up)
Start	All points in one cluster	Each point as a single cluster
Operation	Repeatedly split clusters	Repeatedly merge clusters
End	Each point is a cluster (or desired number of clusters)	All points in one cluster (or desired number of clusters)
Complexity	Higher	Lower (more commonly used)

Applications



Market segmentation



Document classification



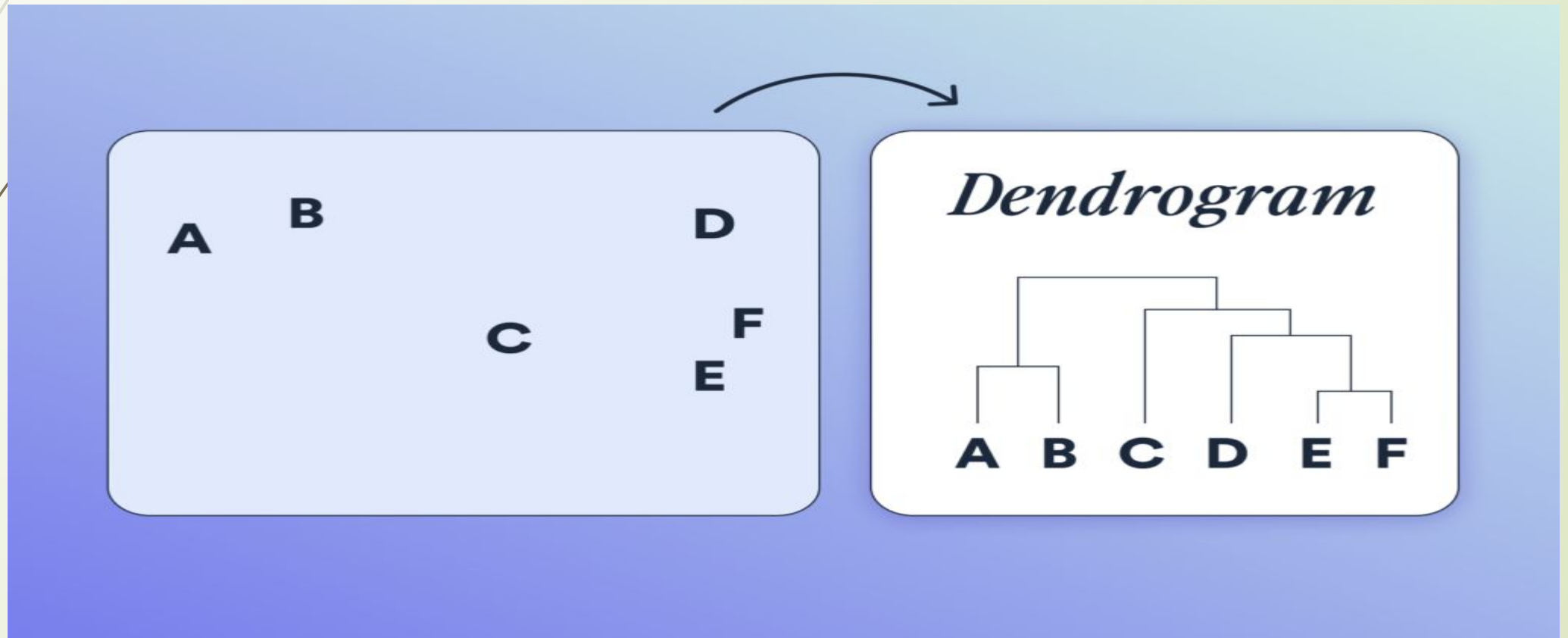
Gene expression analysis



Social network analysis

Dendrogram

A **dendrogram** is a tree-like diagram that shows how clusters are merged.

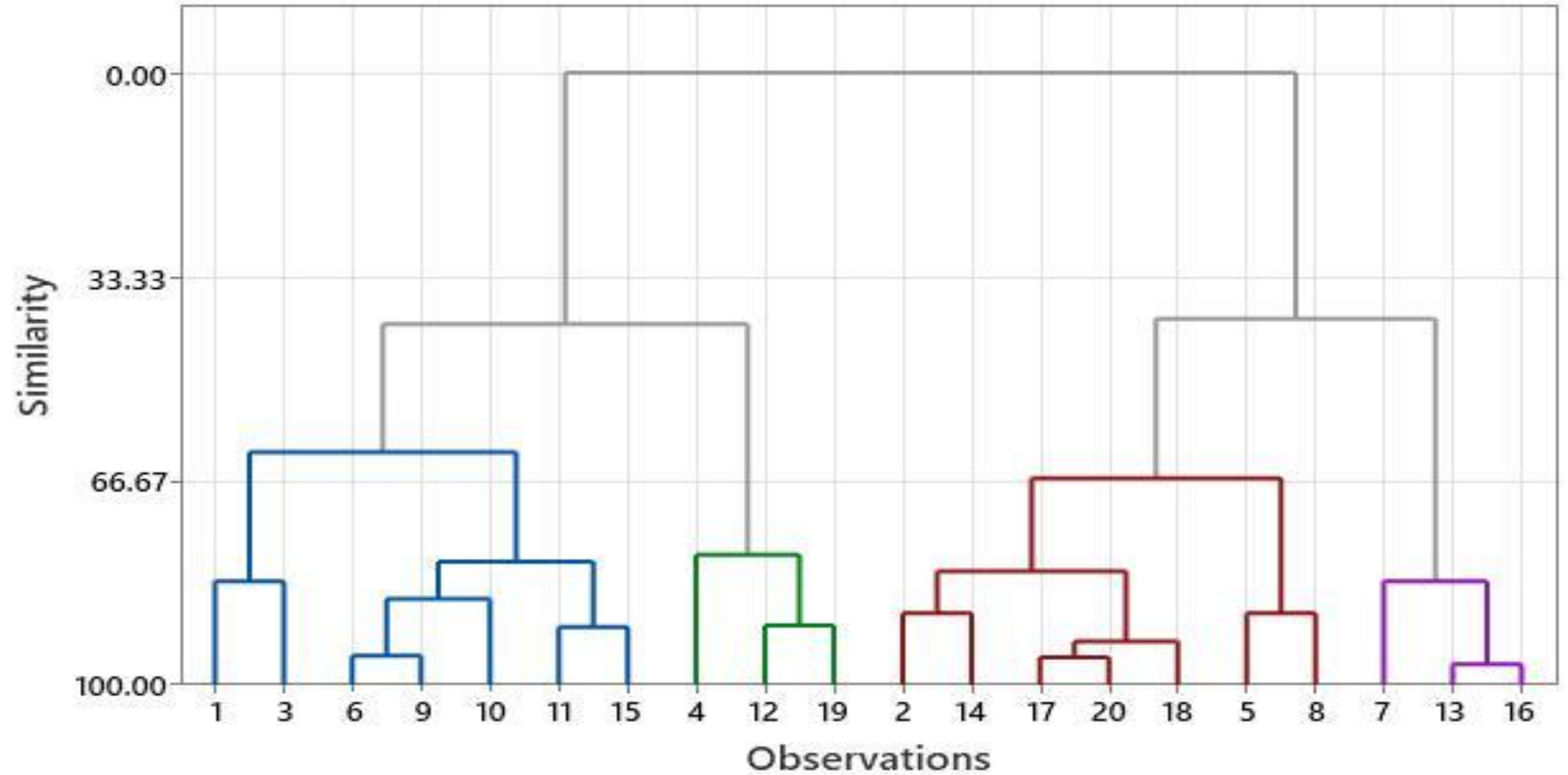


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- ❑ The dendrogram is a tree diagram that displays the groups that are formed by clustering observations at each step and their similarity levels.
 - ❑ The similarity level is measured along the vertical axis (alternately, you can display the distance level), and the different observations are listed along the horizontal axis.

Interpretation

1. Use the dendrogram to view how the clusters are formed at each step and to assess the similarity (or distance) levels of the clusters that are formed.
2. To view the similarity (or distance) levels, hold your pointer over a horizontal line in the dendrogram. The pattern of how similarity or distance values change from step to step can help you to choose the final grouping for your data.
3. The step where the values change abruptly may identify a good point to define the final grouping.
4. The decision about final grouping is also called cutting the dendrogram. Cutting the dendrogram is similar to drawing a line across the dendrogram to specify the final grouping.
5. You can also compare dendrograms for different final groupings to determine which final grouping makes the most sense for your data.

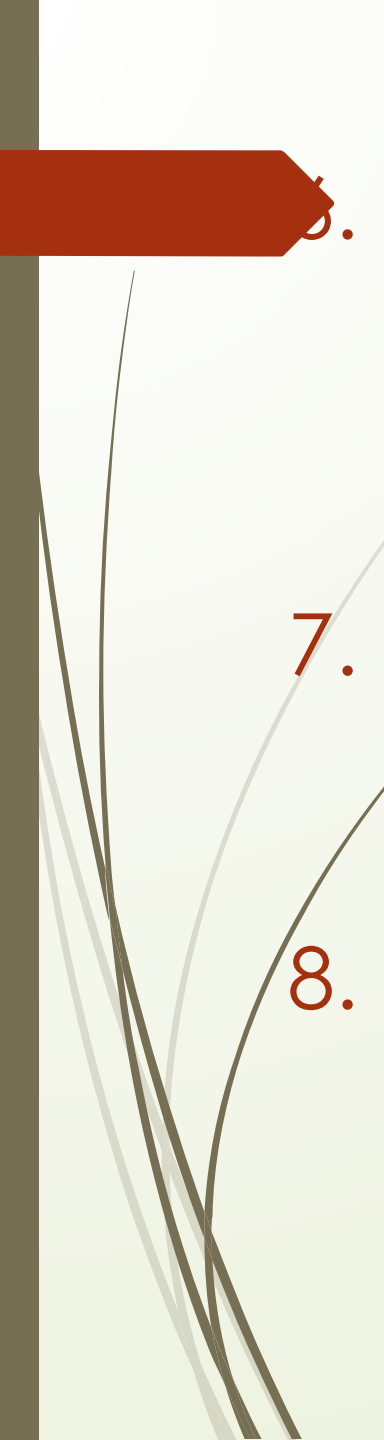
Dendrogram
Complete Linkage, Euclidean Distance





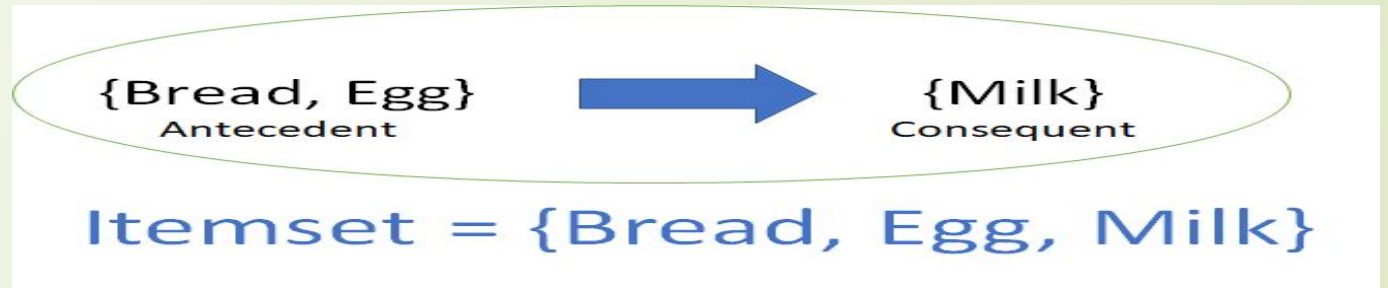
This dendrogram was created using a final partition of 4 clusters, which occurs at a similarity level of approximately 40.

2. The first cluster (far left) is composed of seven observations (the observations in rows 1, 3, 6, 9, 10, 11, and 15 of the worksheet).
3. The second cluster, directly to the right, is composed of 3 observations (the observations in rows 4, 12, and 19 in the worksheet).

- 
6. The third cluster is composed of 7 observations (the observations in rows 2, 14, 17, 20, 18, 5, and 8). The fourth cluster, on the far right, is composed of 3 observations (the observations in rows 7, 13, and 16).
 7. If you cut the dendrogram higher, then there would be fewer final clusters, but their similarity level would be lower.
 8. If you cut the dendrogram lower, then the similarity level would be higher, but there would be more final clusters.

Apriori algorithm for association rule learning problems

- Association rule learning is a **rule-based machine learning** method to find **relationships (associations) between variables** in large datasets.
- **Real life example:**
- If a customer buys **bread** and **butter**, they are likely to buy **jam**.
This is an example of a **market basket analysis**.

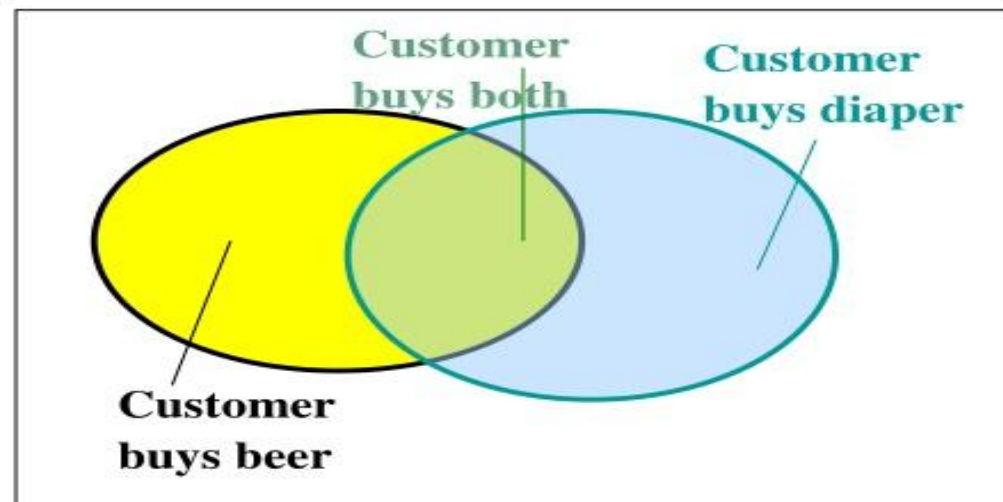


Steps of Apriori Algorithm

- ❑ **Set minimum support and confidence**
- ❑ **Generate all frequent itemsets:**
 - ❑ Count itemsets in the dataset
 - ❑ Eliminate itemsets below the support threshold
- ❑ **Generate association rules** from the frequent item sets
- ❑ **Filter rules** based on confidence and lift

Basic Concepts: Frequent Patterns and Association Rules

Transaction-id	Items bought
10	A, B, D
20	A, C, D
30	A, D, E
40	B, E, F
50	B, C, D, E, F



- Itemset $X = \{x_1, \dots, x_k\}$
- Find all the rules $X \Rightarrow Y$ with minimum support and confidence
 - **support**, s , probability that a transaction contains $X \cup Y$
 - **confidence**, c , conditional probability that a transaction having X also contains Y

Let $sup_{min} = 50\%$, $conf_{min} = 50\%$

Freq. Pat.: $\{A:3, B:3, D:4, E:3, AD:3\}$

Association rules:

$A \Rightarrow D$ (60%, 100%)

$D \Rightarrow A$ (60%, 75%)

APRIORI ALGORITHM EXAMPLE

Database D
Minsup = 0.5

TID	Items
100	1 3 4
200	2 3 5
300	1 2 3 5
400	2 5

Scan D

C_1

itemset	sup.
{1}	2
{2}	3
{3}	3
{4}	1
{5}	3

L_1

itemset	sup.
{1}	2
{2}	3
{3}	3
{5}	3

L_2

itemset	sup
{1 3}	2
{2 3}	2
{2 5}	3
{3 5}	2

C_2

itemset	sup
{1 2}	1
{1 3}	2
{1 5}	1
{2 3}	2
{2 5}	3
{3 5}	2

Scan D

C_2

itemset
{1 2}
{1 3}
{1 5}
{2 3}
{2 5}
{3 5}

C_3

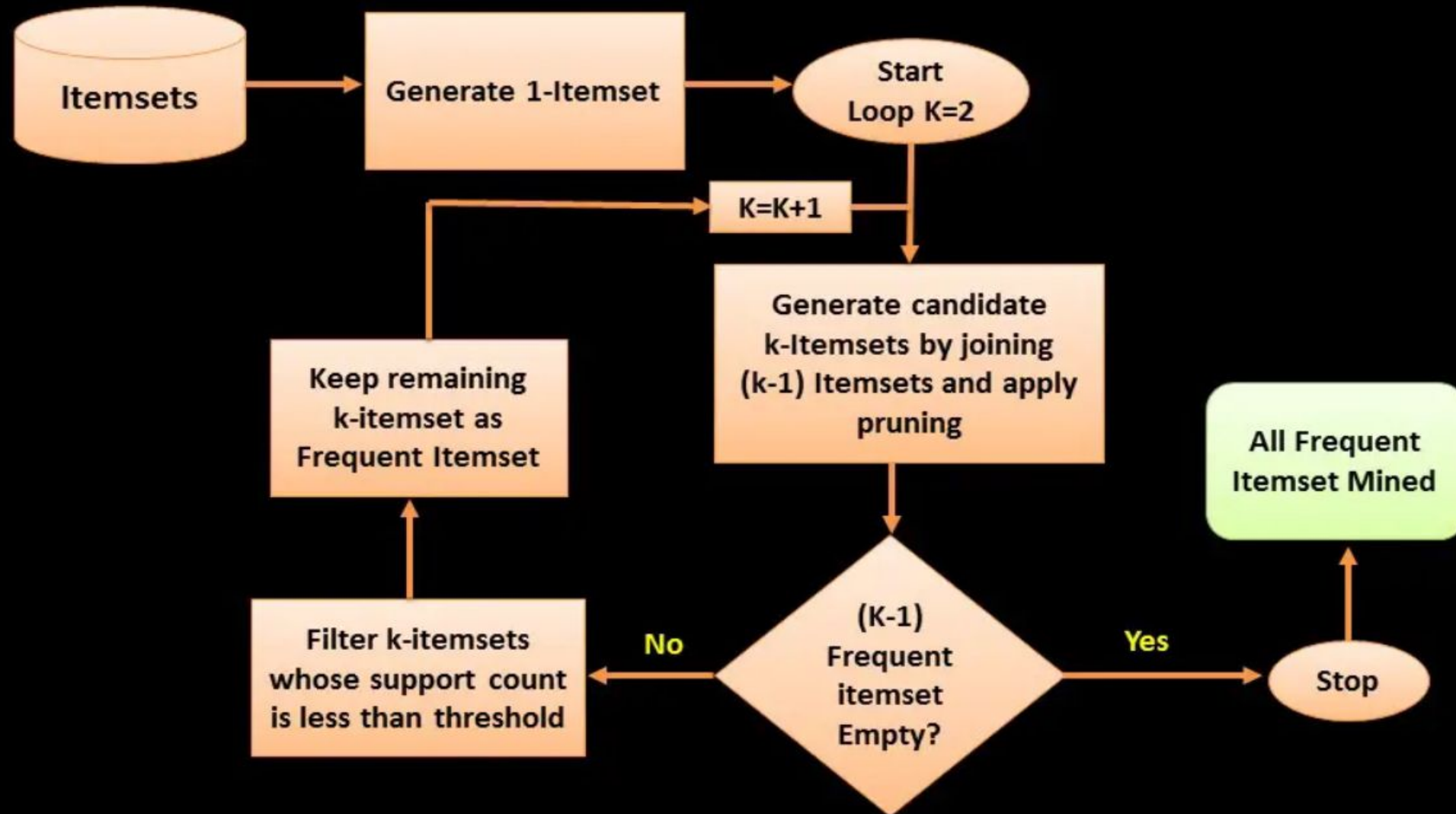
itemset
{2 3 5}

Scan D

L_3

itemset	sup
{2 3 5}	2

How Apriori Algorithm model works



Applications of Apriori Algorithm

- ❑ **E-commerce:** Used to recommend products that are often bought together like laptop + laptop bag, increasing sales.
- ❑ **Food Delivery Services:** Identifies popular combos such as burger + fries, to offer combo deals to customers.
- ❑ **Streaming Services:** Recommends related movies or shows based on what users often watch together like action + superhero movies.

- 
- ❑ **Financial Services:** Analyzes spending habits to suggest personalized offers such as credit card deals based on frequent purchases.
 - ❑ **Travel & Hospitality:** Creates travel packages like flight + hotel by finding commonly purchased services together.
 - ❑ **Health & Fitness:** Suggests workout plans or supplements based on users' past activities like protein shakes + workouts.

Discriminant function

- A **discriminant function** is a function used in **pattern recognition** and **machine learning** to **classify** data points into different classes.
- It evaluates a **feature vector** and assigns it to one of the predefined classes by comparing the **function values**.

Discriminant Functions

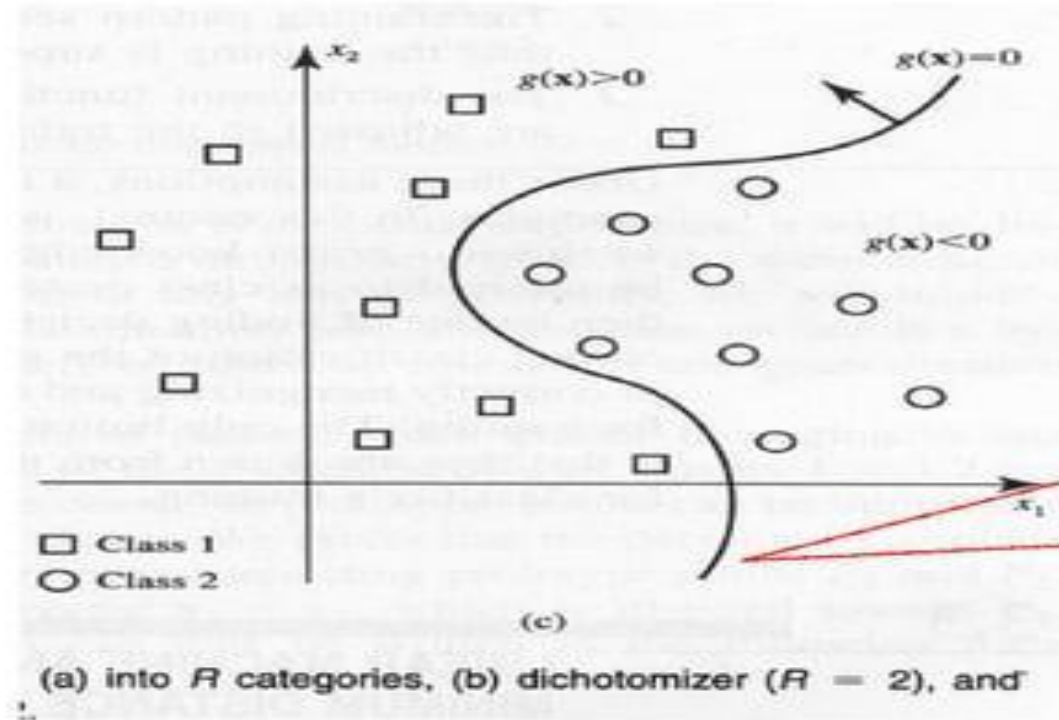
- Although we have focused on probability distribution functions, the decision on class membership in our classifiers has been based solely on the **relative sizes of the probabilities**.
- This observation allows us to *reformulate the classification process in terms of a set of discriminant functions* $y_1(x), \dots, y_c(x)$ such that an input vector $\mathbf{x}$ is assigned to class C_k if:

$$y_k(\mathbf{x}) > y_j(\mathbf{x}) \quad \text{for all } j \neq k.$$

- We can recast the decision rule for minimizing the probability of misclassification in terms of discriminant functions, by choosing:

$$y_k(\mathbf{x}) = P(C_k|\mathbf{x}).$$

Discriminant Functions...



The design of discriminator for this case is not straightforward. The discriminant functions may result as nonlinear functions of x_1 and x_2

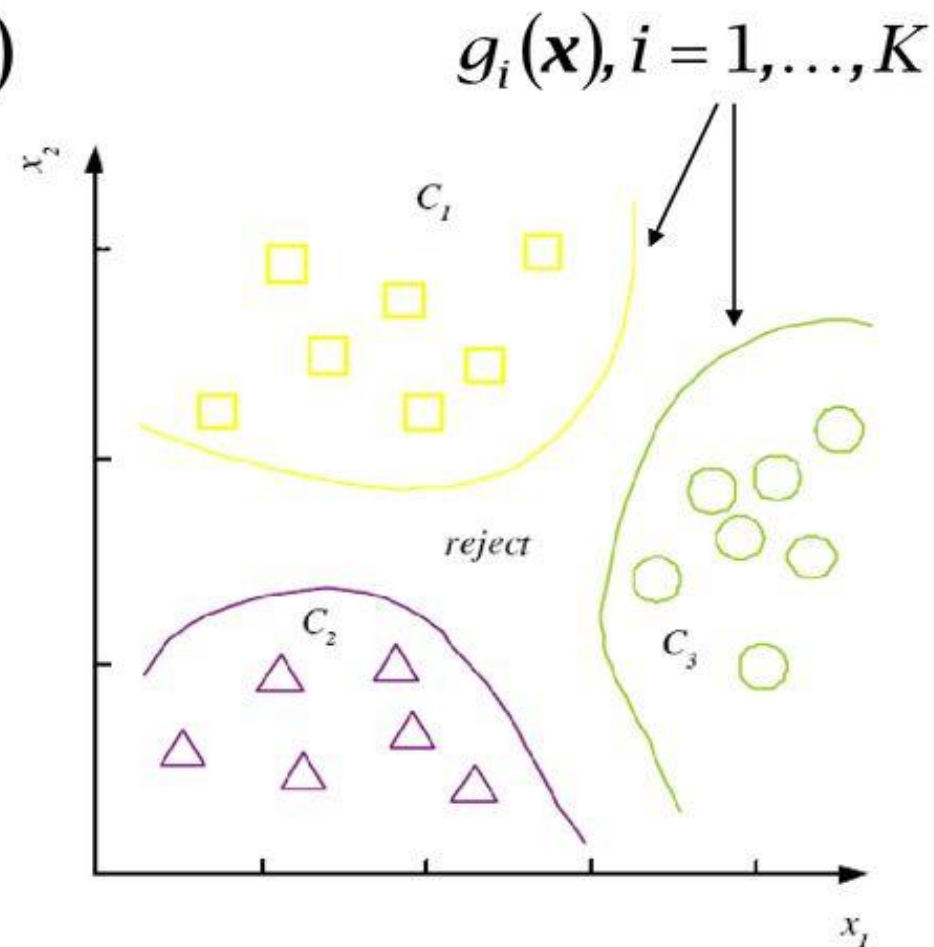
Discriminant Functions

choose C_i if $g_i(\mathbf{x}) = \max_k g_k(\mathbf{x})$

$$g_i(\mathbf{x}) = \begin{cases} -R(\alpha_i | \mathbf{x}) \\ P(C_i | \mathbf{x}) \\ p(\mathbf{x} | C_i)P(C_i) \end{cases}$$

K decision regions $\mathcal{R}_1, \dots, \mathcal{R}_K$

$$\mathcal{R}_i = \{\mathbf{x} \mid g_i(\mathbf{x}) = \max_k g_k(\mathbf{x})\}$$





Why Use Discriminant Functions?

- They provide a **decision boundary** between classes.
- Useful in **supervised learning** for classification tasks.
- Allow for **probabilistic** or **distance-based** interpretations



Types of Discriminant Functions :

- ❑ Linear Discriminant Function (LDF)**
- ❑ Quadratic Discriminant Function (QDF)**
- ❑ Bayesian Discriminant Function (BDF)**



Applications Of Discriminant Functions:

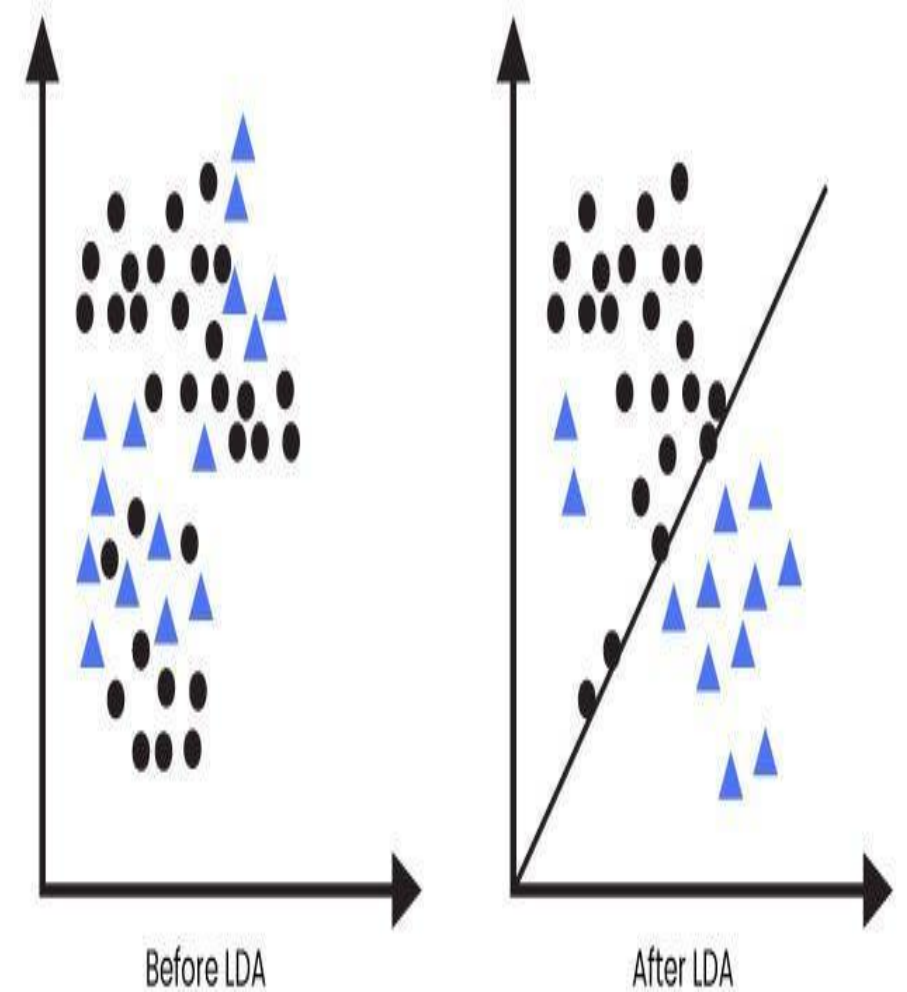
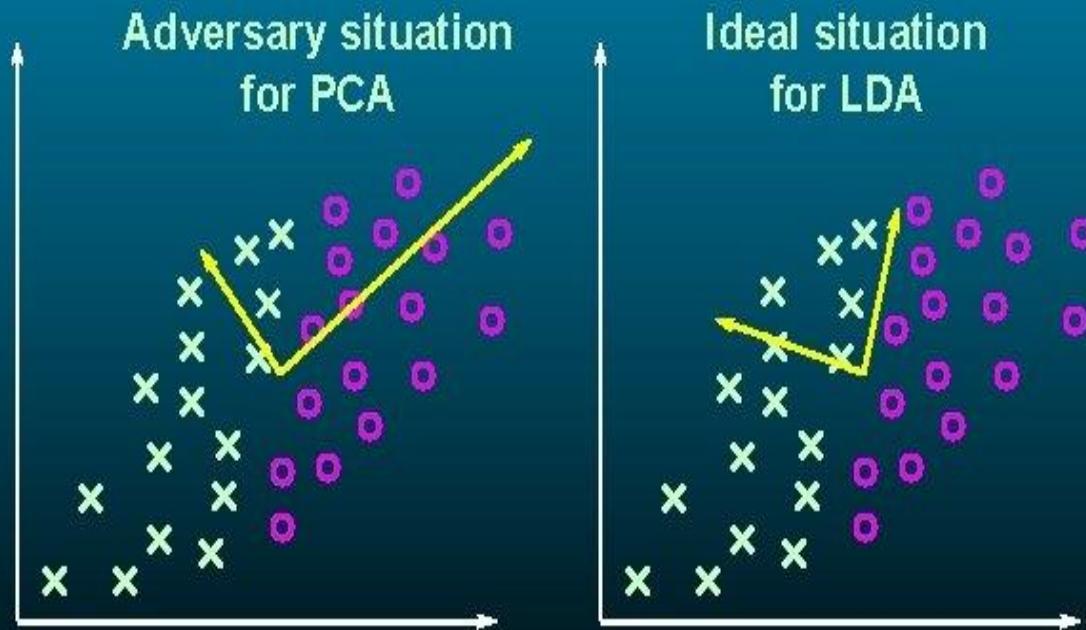
- ❑ Face recognition
- ❑ Handwriting digit recognition
- ❑ Medical diagnosis
- ❑ Spam detection

Linear Discriminant Function:

- Linear discriminant analysis (LDA) is a **supervised learning algorithm** used for classification and dimensionality reduction in machine learning.
- It aims to find a linear combination of features that best **separates different classes** in a dataset.

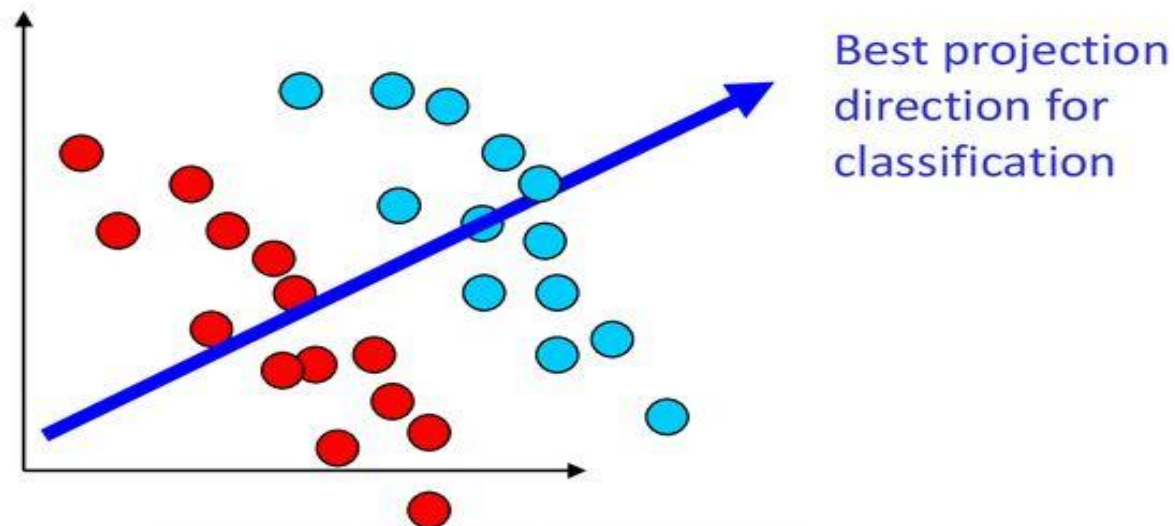
Linear Discriminant Analysis

LDA projection onto directions that can best separate data of different classes.



Linear Method: Linear Discriminant Analysis (LDA)

- LDA finds the projection that best separates the two classes
- Multiple discriminant analysis (MDA) extends LDA to multiple classes



Hypothesis Space

What is a Hypothesis?

- In Machine Learning, a **hypothesis** is a mathematical function or model that maps input features (**X**) to an output (**Y**).
- It is the model's prediction function learned from the training data.
- Mathematically,

$$h : X \rightarrow Y$$

where:

- **X** = Input (Features)
- **Y** = Output (Target/Class)
- **h** = Hypothesis

Example:

Suppose we want to predict the marks of a student based on study hours.

Study Hours	Marks
2	35
4	55
6	72

A possible hypothesis is:

$$h(x) = 10x + 15$$

Where,

- x = Study Hours
- $h(x)$ = Predicted Marks

What is Hypothesis Space?

- The Hypothesis Space is the set of all possible hypotheses (models) that a learning algorithm can choose from to solve a problem.
- It is denoted by

$$H = \{h_1, h_2, h_3, \dots, h_n\}$$

where each hypothesis represents a different model.

Hypothesis Space is the collection of all candidate functions that can approximate **the relationship between inputs and outputs.**

Why is Hypothesis Space Needed?

1. A machine learning algorithm does **not know** the correct model initially.
2. Instead, it searches among many possible models to find the one that best fits the training data.
3. The collection of all these possible models is called the **Hypothesis Space.**

Real-Life Analogy

"Find the best route from your home to college."

Possible routes are:

1.Route A 2. Route B 3. Route C 4. Route D

Each route is a possible solution.

Similarly,

1. Each model = one hypothesis
2. All models together = hypothesis space
3. The ML algorithm chooses the best route (best hypothesis).

Example of Hypothesis Space

Suppose we want to classify emails as Spam or Not Spam.

Possible hypotheses:

1. Hypothesis 1

If email contains "Lottery"

→ Spam

2. Hypothesis 2

If email contains "Free" → Spam

3. Hypothesis 3

If email contains "Lottery" AND "Prize"

→ Spam

4. Hypothesis 4

Always predict Not Spam

All these hypothesis together form Hypothesis SPACE

Mathematical Representation

Suppose,

$$Y = wx + b$$

Different values of **w** and **b** produce different hypotheses.

$$h_1(x) = 2x + 1$$

$$h_2(x) = 3x + 5$$

$$h_3(x) = 5x - 2$$

The collection

$$H = \{2x + 1, 3x + 5, 5x - 2, \dots\}$$

is the Hypothesis Space.

VC Dimension (Vapnik–Chervonenkis Dimension)

1. VC dimension measures the **capacity (complexity)** of a hypothesis space — how well a model can fit various patterns.
2. It is the **maximum number of points** that can be **shattered** (i.e., classified correctly in all possible ways) by the hypothesis class.

□ Examples:

- VC Dimension of a **linear classifier in 2D**: 3
- VC Dimension of a **decision stump (1-level decision tree)**: 1
- VC Dimension of a **k-nearest neighbor** (if $k=1$): **infinite**

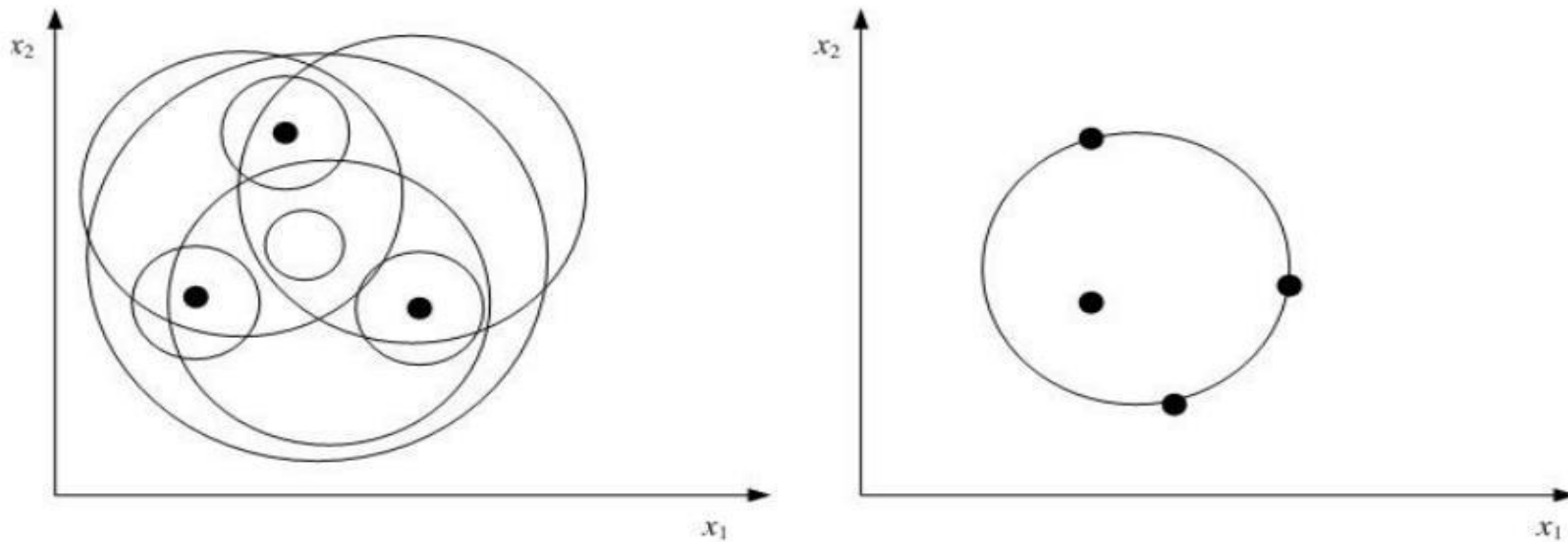
VC Dimension

dimension

- ▶ Vapnik-Chervonenkis $VC(H)$: the size of the largest finite subset of X shattered by H .
- ▶ If $VC(H) = k$, then for all $k + 1$ points, there exists a labeling that cannot be shattered
 - ▶ can't find a hypothesis in H consistent with it
- ▶ If arbitrarily large finite sets of X can be shattered by H then $VC(H) = \infty$

VC-dimension

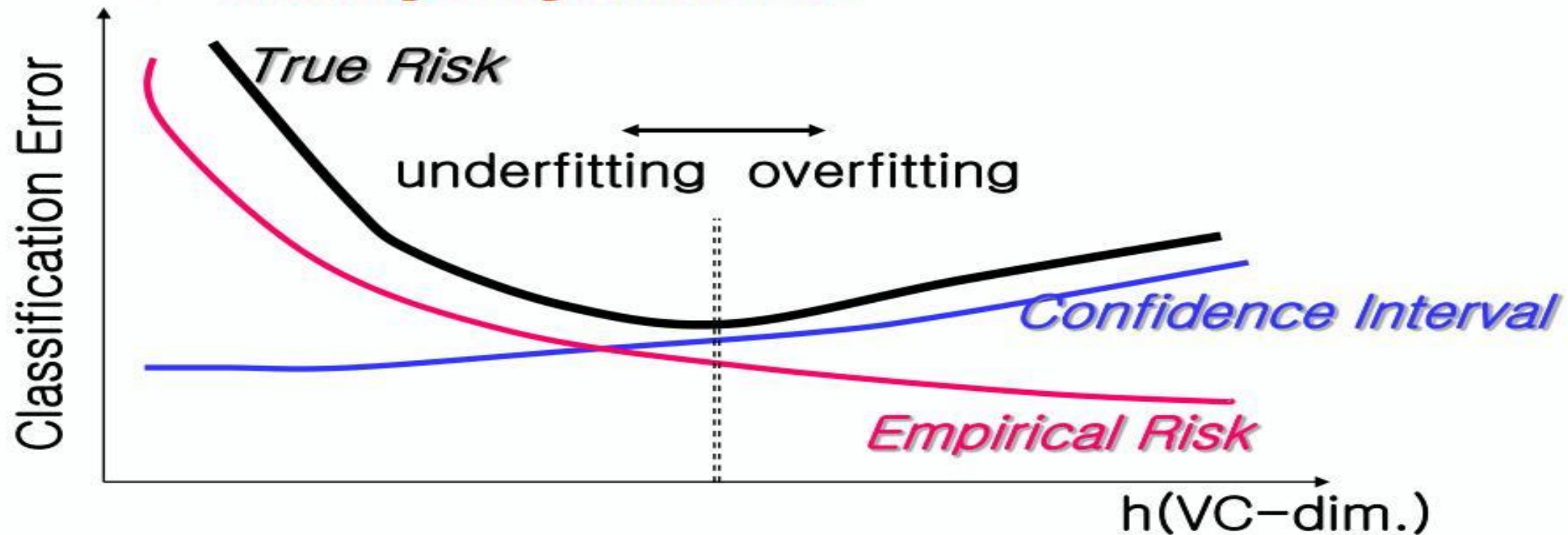
- **Example:** spherical decision functions $f(c, r, \mathbf{x})$ can shatter 3 points *BUT cannot shatter* 4 points



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- The Vapnik-Chervonenkis (VC) dimension is a measure of the capacity of a hypothesis set **to fit different data sets**.
 - It was introduced by Vladimir Vapnik and Alexey Chervonenkis in the 1970s and has become a fundamental concept in statistical learning theory.
 - The VC dimension is a measure of the **complexity of a model**, which can help us understand how well it can fit different data sets.
 - The VC dimension of a hypothesis set **H is the largest number** of points that can be shattered by H.
 - A hypothesis set H shatters a set of points S if, for every possible labeling of the points in S, there exists a hypothesis in H that correctly classifies the points

VC Dimension and Empirical Risk

- Empirical Risk is Decreasing Function of VC Dimension
 - Need a principled methods for the minimization



Why is it

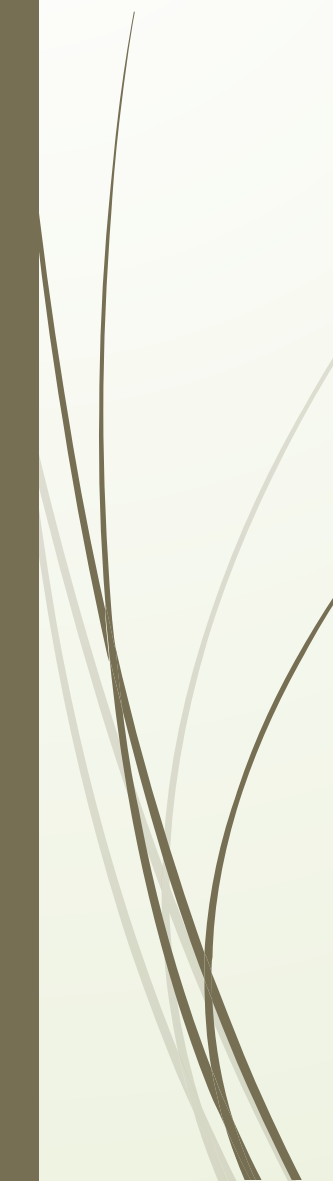
important?

- Helps understand underfitting vs overfitting
- A model with high VC dimension can overfit.
- A balance is needed between model complexity and generalization



3.3

Ensemble Methods (Boosting, Bagging, Random Forests)



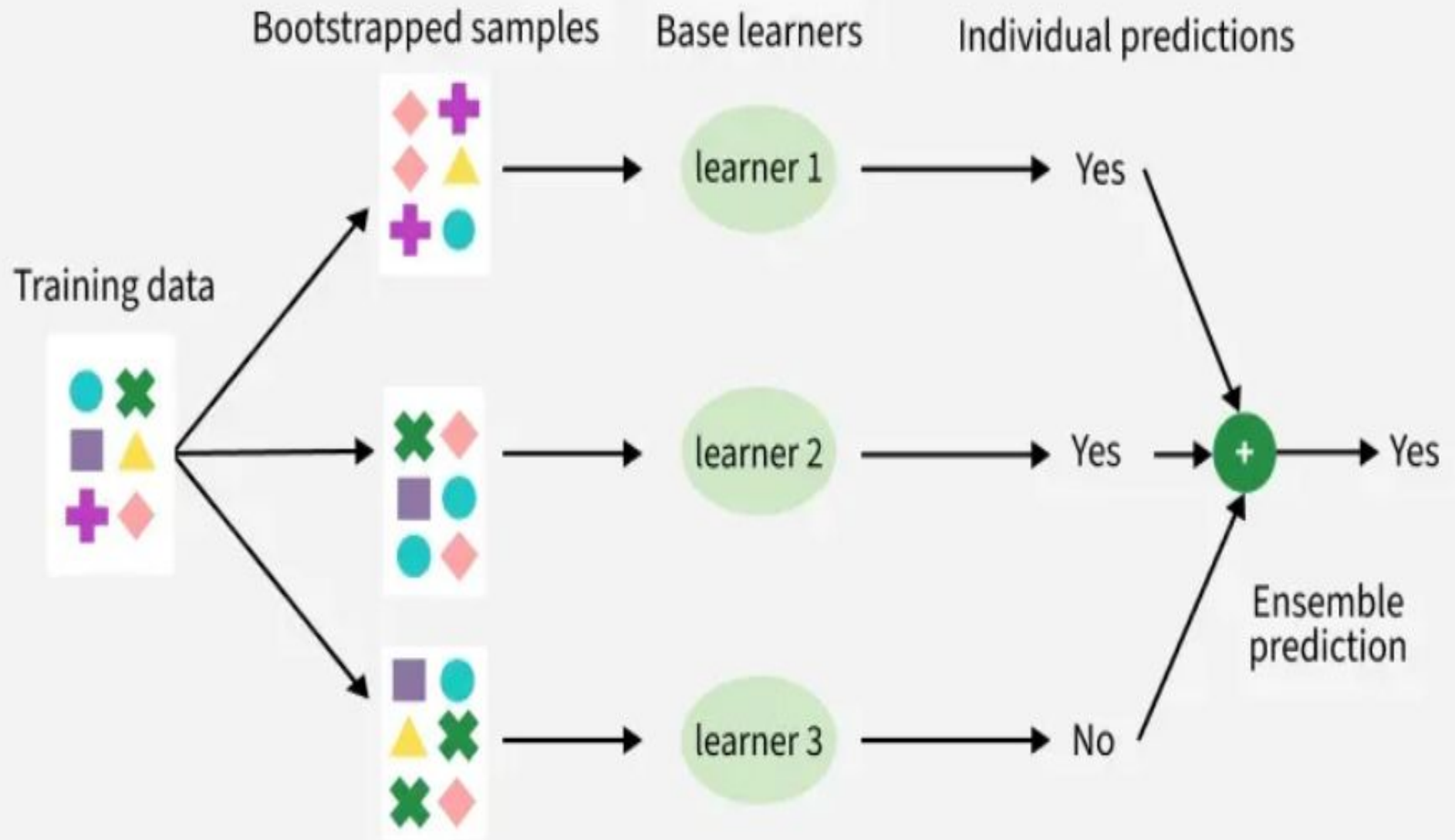
Ensemble learning

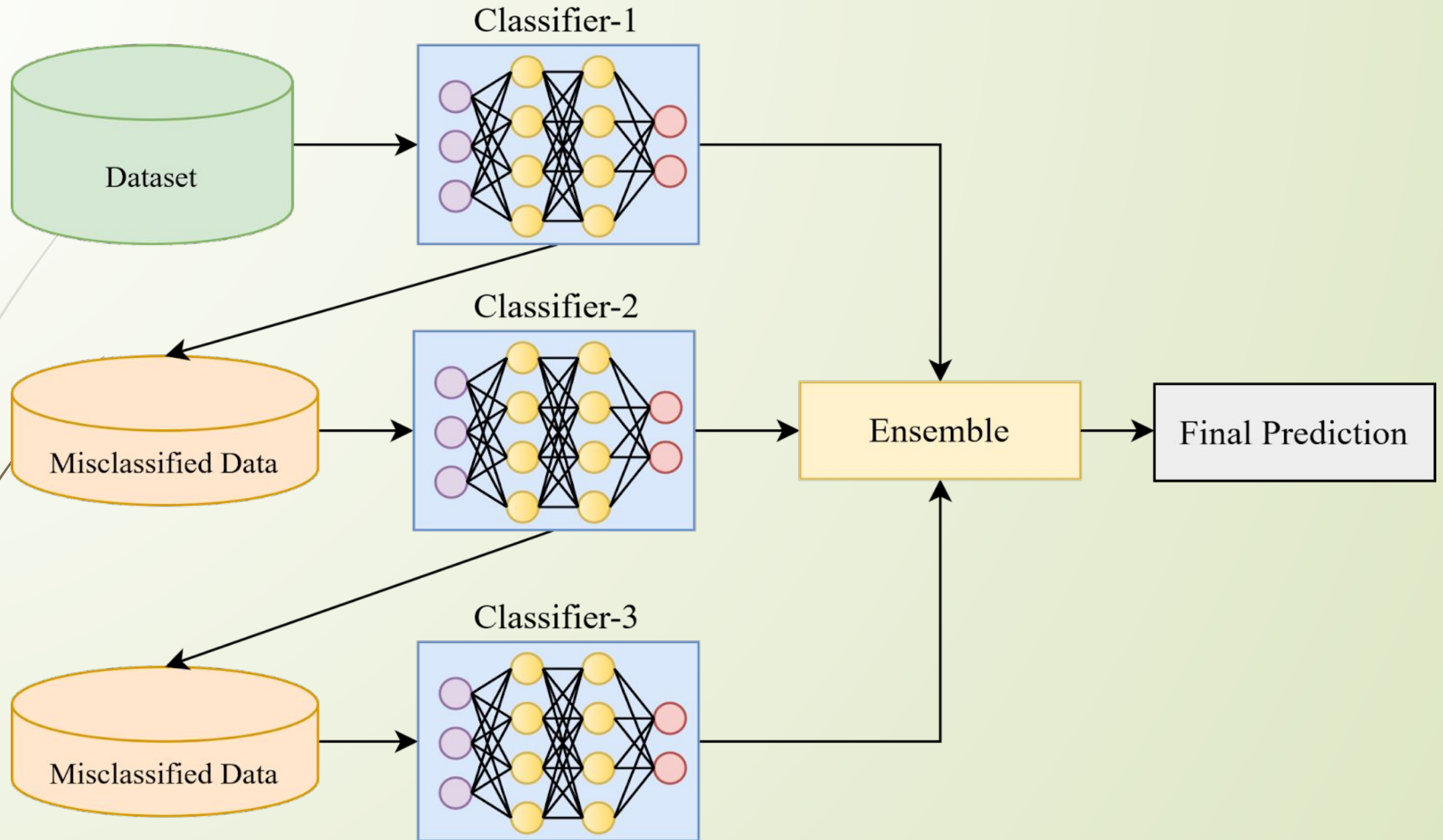
- **Ensemble learning** is a technique in machine learning where **multiple models (often called "weak learners")** are trained and combined to solve the same problem.
- The idea is that **a group of models working together can outperform a single strong model.**

When to Use Ensemble Learning?

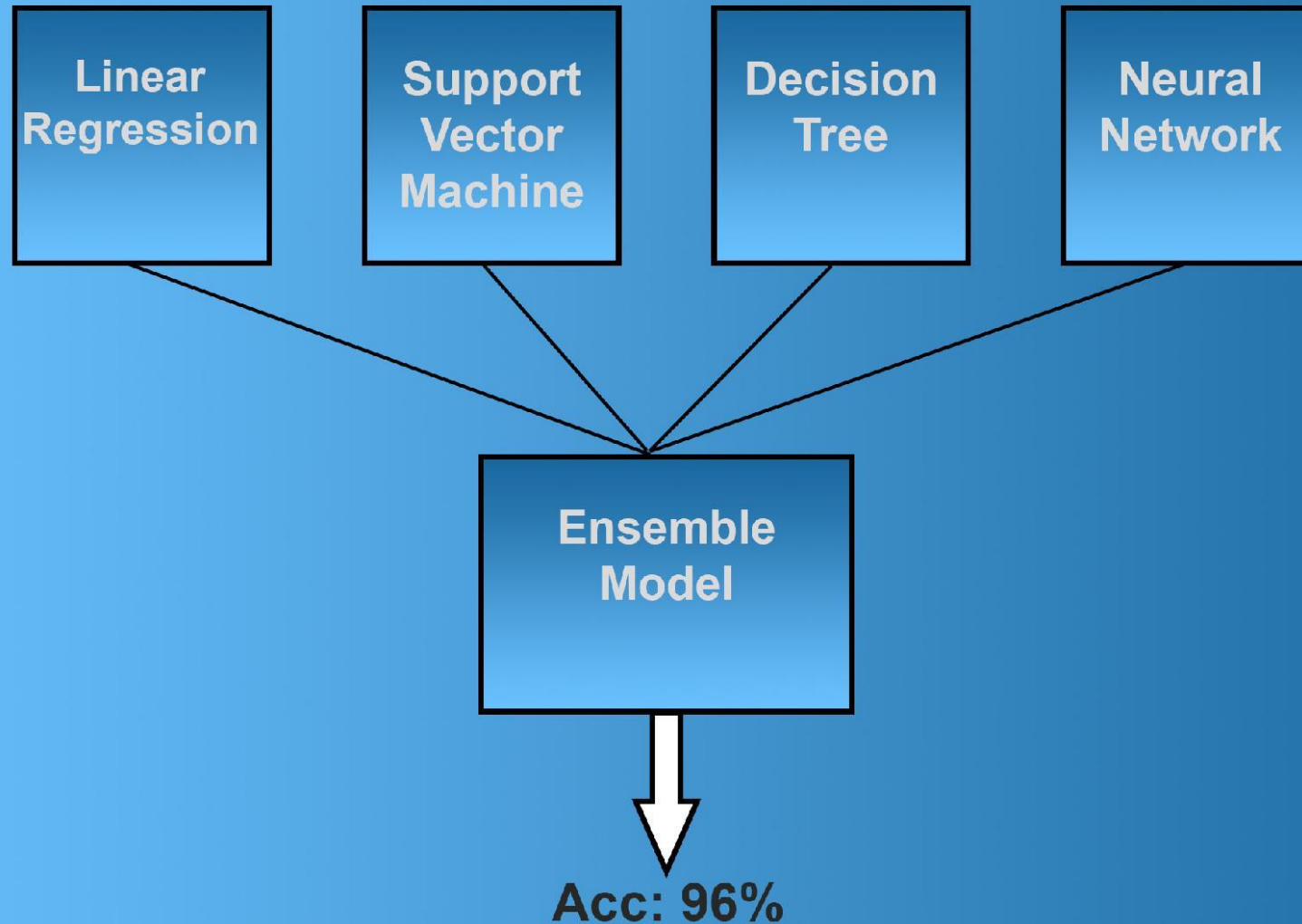
- ❑ You have **high variance or high bias** in your model.
- ❑ Your base models are **diverse and complementary**.
- ❑ You want to **increase model performance** in competitions (e.g., Kaggle).
- ❑ **Real-Life Example**
 - ❑ Imagine trying to guess a movie's rating:
 - ❑ One friend uses past ratings
 - ❑ Another reads online reviews
 - ❑ A third watches the trailer
 - Each has weaknesses alone, but combining their opinions gives a better estimate. That's ensemble learning!

Ensemble learning

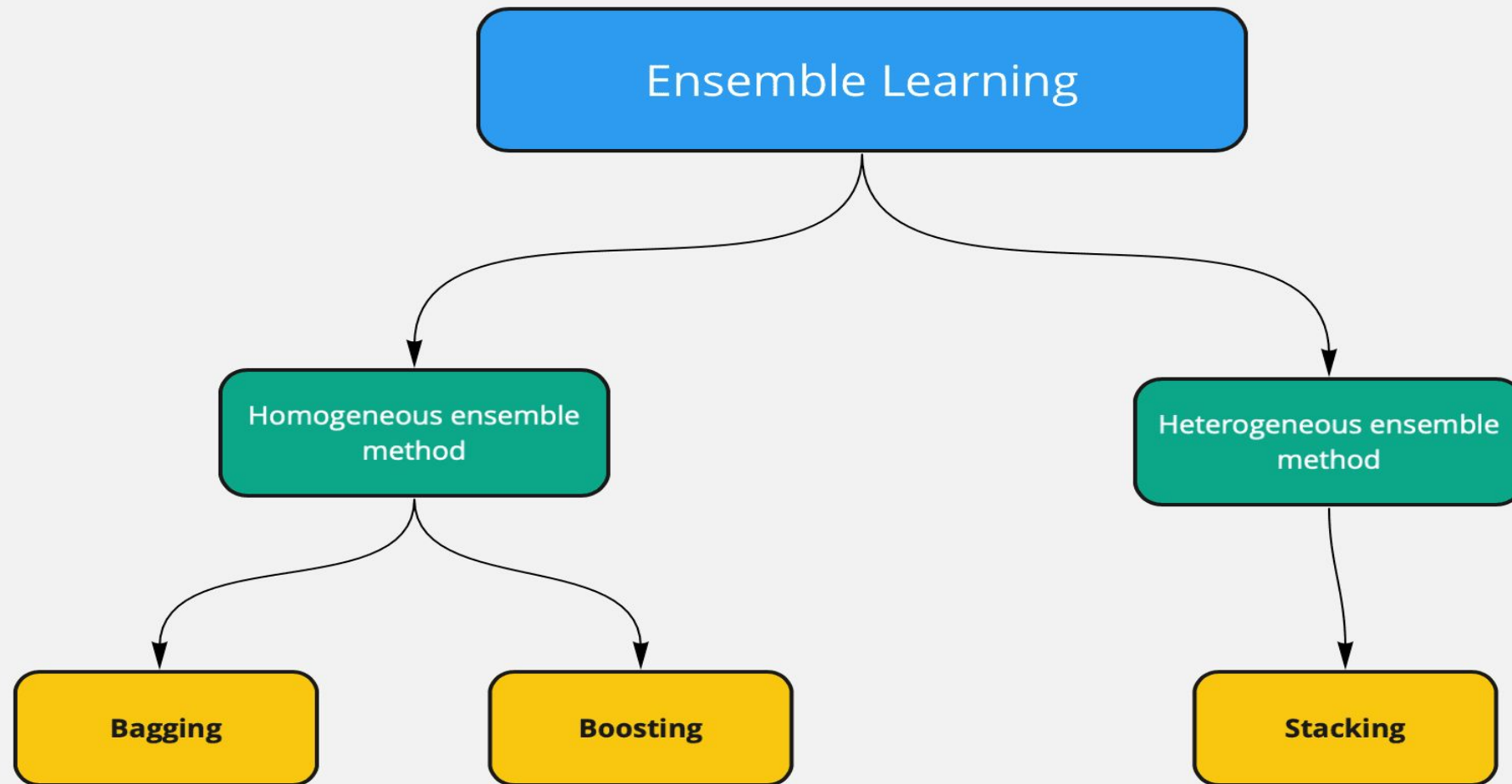




Accuracy: Highest



Types of Ensemble learning



❑ **Bagging (Bootstrap Aggregating):**

Models are trained independently on different random subsets of the training data.

❑ Their results are then combined—usually by averaging (for regression) or voting (for classification).

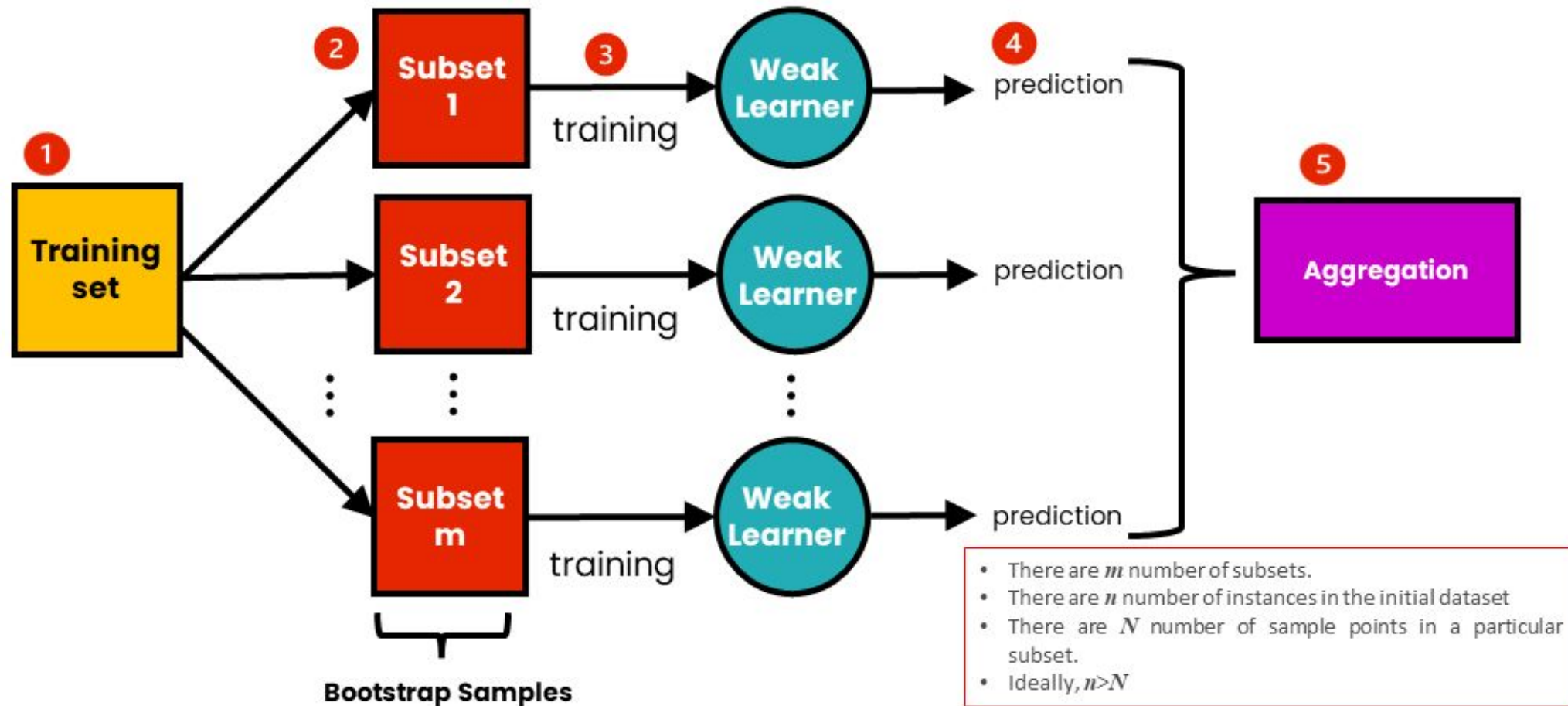
❑ This helps reduce variance and prevents over fitting.

❑ **Example Algorithms:**

❑ **Random Forest** (uses decision trees + bagging)

How Bagging Learning model works:

The Process of Bagging (Bootstrap Aggregation)

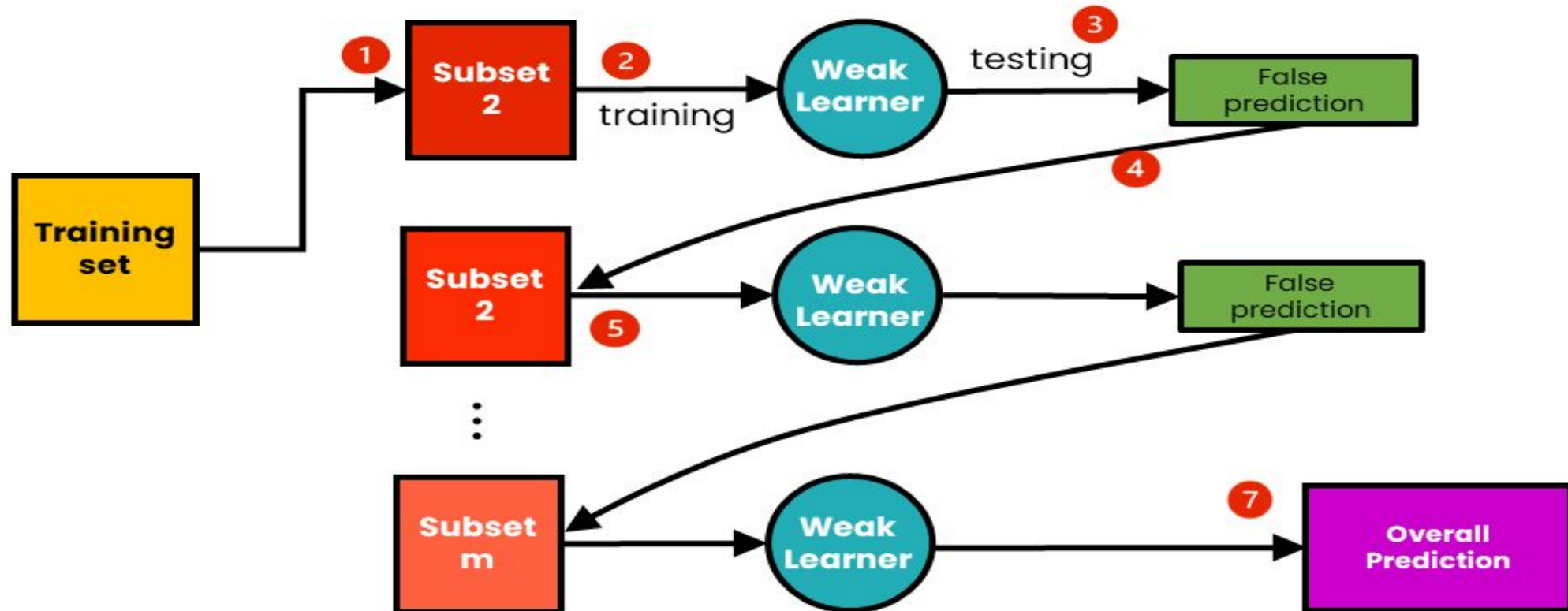


Boosting:

Models are trained one after another. Each new model focuses on fixing the errors made by the **previous ones**.

The final prediction is a weighted combination of all models, which helps **reduce bias and improve accuracy**.

The Process of Boosting





Models are trained **sequentially**, each new model focusing on **correcting the errors** of the previous ones.

□ Final prediction is a **weighted sum** of all models.

□ **Reduces bias and variance**

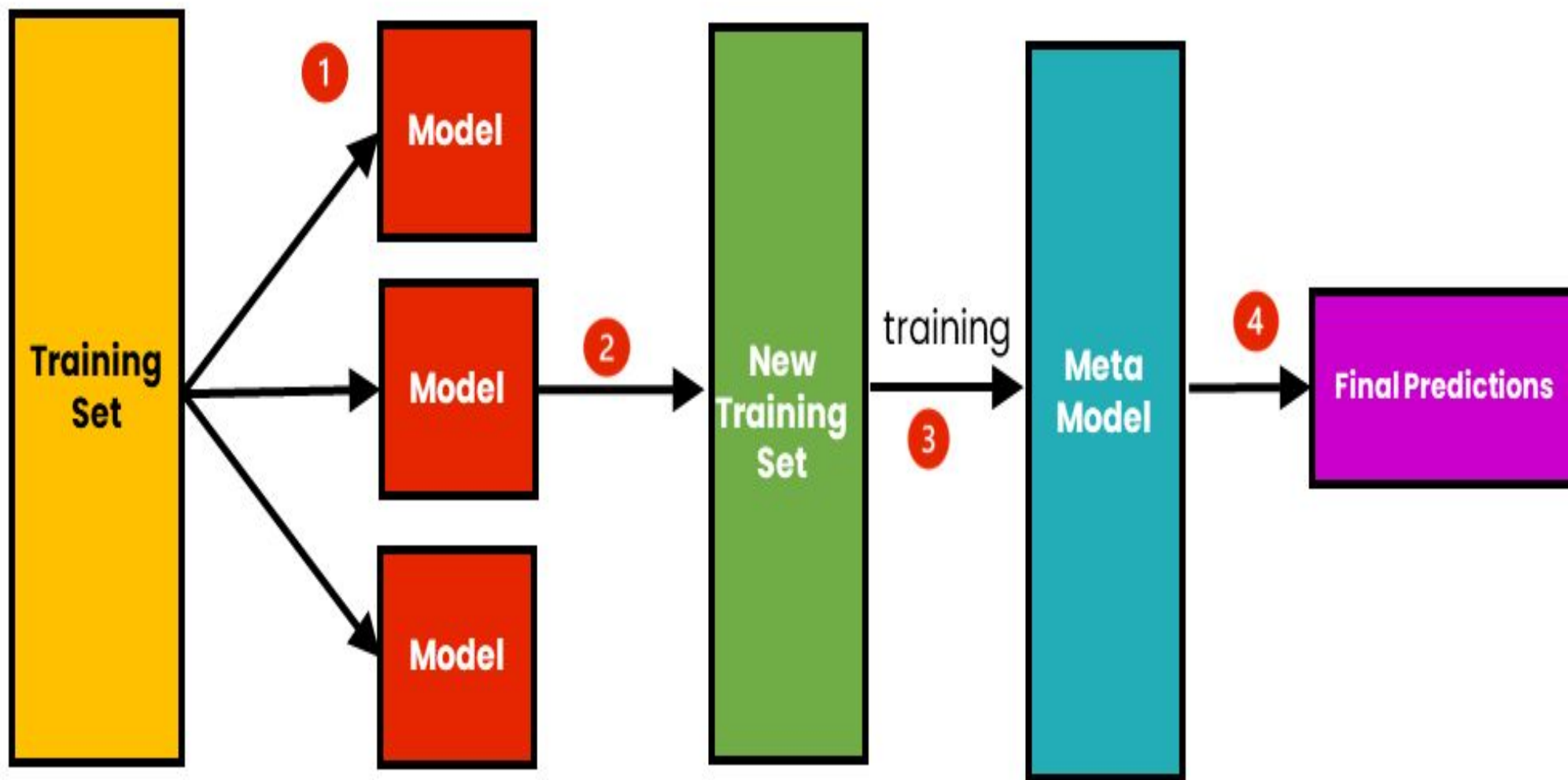
□ **Popular Boosting Algorithms:**

- ✓ AdaBoost (Adaptive Boosting)
- ✓ Gradient Boosting (GBM)
- ✓ XGBoost
- ✓ LightGBM
- ✓ CatBoost

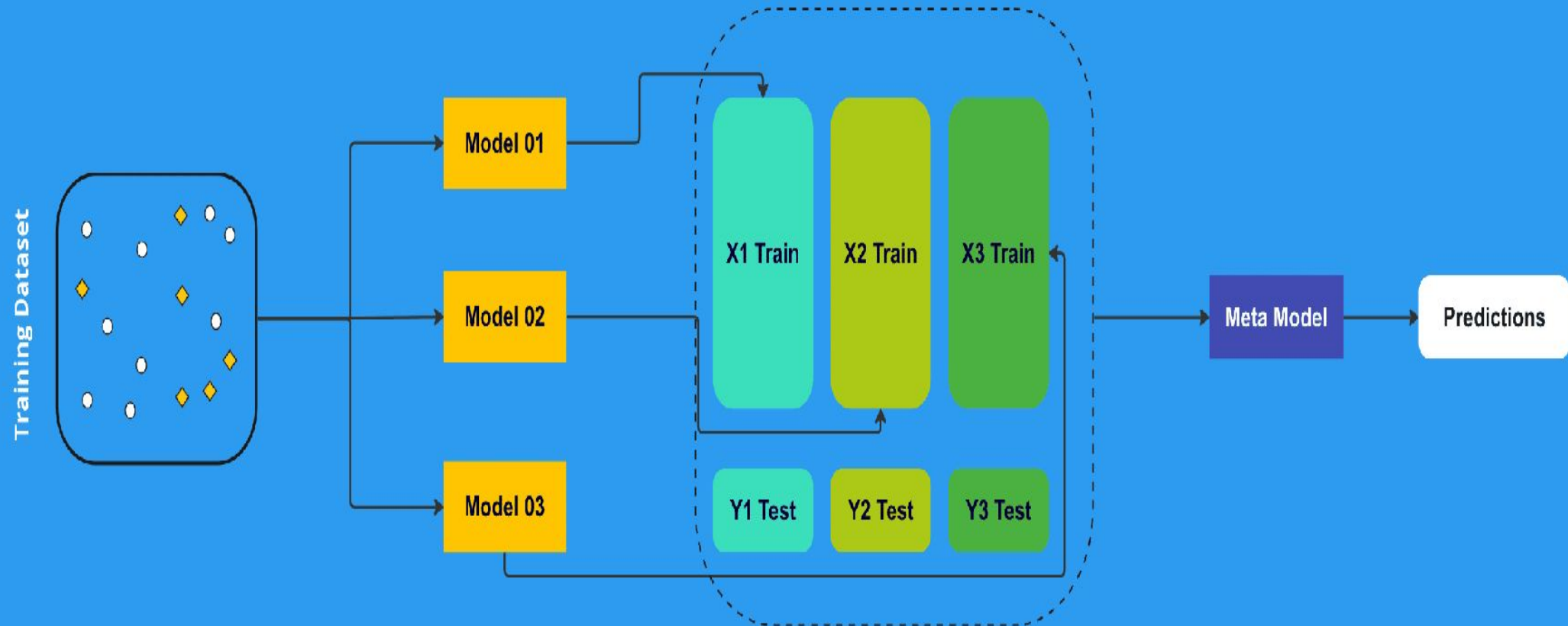
Stacking (Stacked Generalization)

- ❑ Multiple different models (often of different types) are trained, and their predictions are used as inputs to a final model, called a meta-model.
- ❑ The meta-model learns how to best combine the predictions of the base models, aiming for better performance than any individual model.
- ❑ The predictions of base models are fed to a **meta-model** (e.g., logistic regression) that learns how to best combine them.
- ❑ **Leverages strengths of different models**
- ❑ Useful when base learners are diverse.

The Process of Stacking



How Stacking Technique Improves Machine Learning Model's Performance



Model Dataset - - - - -> Level 01 Modeling - - - - -> Level 01 Predictions - - - - -> Level 02 Modeling - - - - -> Final Predictions

Real-Life Example

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